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ORIGINAL RESEARCH

Ultrasound-Guided Low-Level He-Ne Laser Therapy Following Acupotomy Release for Carpal Tunnel Syndrome: A Multicenter Retrospective Study

Xiang Shang[1,*], Guo-Rui Luan[2,*], De-Hong Meng[3], Fen Zhang[3], San-Bing Wu[3], Hou-Shan Fang[4], Fei Li[3], Yong-Hui Yang[3]

[1]Department of First Clinical Medical College, Anhui University of Chinese Medicine, Hefei, People's Republic of China; [2]Department of Tuina, Shuguang Anhui Hospital Affiliated to Shanghai University of Traditional Chinese Medicine, Hefei, People's Republic of China; [3]Department of Orthopedics, The Second Affiliated Hospital of Anhui University of Chinese Medicine, Hefei, People's Republic of China; [4]Department of Orthopedics, Lu'an Traditional Chinese Medicine Hospital, Luan, People's Republic of China

*These authors contributed equally to this work

Correspondence: Yong-Hui Yang; Fei Li, Department of Orthopedics, The Second Affiliated Hospital of Anhui University of Chinese Medicine, Hefei, People's Republic of China, Email yangyonghui@ahtcm.edu.cn; leagcen@163.com

Background: Carpal tunnel syndrome (CTS) is a common compressive neuropathy associated with pain, sensory disturbance, and functional impairment. Although corticosteroid injection is widely used as an adjunctive treatment for minimally invasive decompression, steroid-related adverse effects limit its clinical application in some patients. This study aimed to evaluate the efficacy and safety of ultrasound- guided low-level He-Ne laser therapy following acupotomy release as a steroid-sparing treatment strategy for CTS.

Methods: This multicenter retrospective study included 163 patients with CTS who underwent one of the following treatments: ultrasound-guided acupotomy combined with low-level He-Ne laser therapy group (ALG, n=54), acupotomy combined with corticosteroid injection group (ACG, n=55), or acupotomy release group (AG, n=54). Pain intensity was assessed using the Visual Analog Scale (VAS), and functional outcomes were evaluated using the Boston Carpal Tunnel Questionnaire (BCTQ). Electrophysiological and ultrasonographic parameters of the median nerve were also analyzed before and after treatment. Adverse events were recorded during follow-up.

Results: All three groups showed significant improvements in pain, functional outcomes, electrophysiological findings, and imaging parameters after treatment ($P<0.05$). Compared with the AG, both the ALG and ACG demonstrated significantly greater reductions in VAS and BCTQ scores at 1, 4, and 16 weeks after treatment ($P<0.05$). No significant differences in VAS scores were observed between the ALG and ACG during follow-up. Improvements in electrophysiological and ultrasonographic outcomes were also more pronounced in the ALG and ACG than in the AG. No serious complications were observed in any group. Steroid-related hyperglycemia occurred in six patients in the ACG, whereas no laser-related adverse events were reported in the ALG.

Conclusion: Ultrasound-guided low-level He-Ne laser therapy combined with acupotomy release significantly improved pain, function, and nerve recovery in CTS patients. Its therapeutic efficacy was comparable to corticosteroid injection while avoiding steroid-related adverse events, suggesting that it may represent a promising steroid-sparing treatment strategy.

Keywords: carpal tunnel syndrome, low-level laser therapy, He-Ne laser, pain management, acupotomy, ultrasound-guided intervention

Introduction

Carpal tunnel syndrome (CTS) is one of the most common compressive peripheral neuropathies and is primarily caused by compression of the median nerve within the carpal tunnel.[1] Patients with CTS commonly present with pain, numbness, paresthesia, and impaired hand function, which can substantially affect daily activities and quality of life.[2–4] The prevalence

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of CTS has increased worldwide, particularly among individuals engaged in repetitive wrist activities and middle-aged women, making it an important public health concern.[5,6]

Current management strategies for CTS include wrist splinting, oral medications, corticosteroid injection, and surgical decompression.[7–9] Among these approaches, corticosteroid injection is widely used because of its short-term anti-inflammatory and analgesic effects.[7] However, steroid-related adverse effects, including hyperglycemia, adrenal suppression, and other metabolic complications, may limit its clinical applicability, particularly in patients with diabetes or metabolic disorders.[10] Surgical decompression remains the definitive treatment for severe CTS, but concerns regarding surgical trauma, scar formation, postoperative pain, and prolonged recovery continue to affect patient acceptance.[11,12] New research has found that splinting and kinesio taping are equally effective in improving pain, hand grip strength, and median nerve cross-sectional area in patients with CTS.[13] Extracorporeal shock wave therapy has also been shown to alleviate pain and paresthesia in this patient population.[14] Therefore, identifying safe and effective minimally invasive steroid-sparing treatment strategies for CTS remains clinically important.

In recent years, ultrasound-guided minimally invasive decompression techniques have attracted increasing attention in the management of CTS. Acupotomy release can mechanically reduce compression within the carpal tunnel by releasing the transverse carpal ligament, thereby improving median nerve entrapment.[15] Conventional blind procedures, however, may increase the risk of neurovascular injury because of limited visualization.[16] Ultrasound guidance enables real-time visualization of anatomical structures and instrument positioning, improving procedural precision and safety.[17] In addition to mechanical decompression, low-level laser therapy has demonstrated anti-inflammatory, analgesic, and tissue repair-promoting effects in various musculoskeletal disorders.[18,19] These properties suggest that low-level He-Ne laser therapy may serve as a potential adjunctive strategy for improving pain and functional recovery after minimally invasive decompression in CTS patients.

Previous literature demonstrated satisfactory therapeutic efficacy of ultrasound-guided acupotomy release combined with corticosteroid injection for carpal tunnel syndrome.[20] Nevertheless, metabolic adverse events related to steroid administration remain an important clinical limitation for vulnerable patient populations. Therefore, the present multi-center retrospective study aimed to evaluate the efficacy and safety of ultrasound-guided low-level He-Ne laser therapy following acupotomy release as a steroid-sparing minimally invasive treatment strategy for CTS.

Participants and Methods

Participants

This multicenter retrospective study included patients diagnosed with carpal tunnel syndrome (CTS) who received treatment at three medical centers between January 2021 and December 2023. The study protocol was approved by the institutional ethics committees of all participating centers, and all procedures were conducted in accordance with the Declaration of Helsinki. Written informed consent for treatment and data collection was obtained from all patients prior to participation. This study was approved by the Ethics Committee of A Hospital (Ethics No.: 2025SGH–EAD–020).

Patients were allocated to one of the following treatment groups according to shared decision-making between physicians and patients based on clinical characteristics and treatment preferences: ultrasound-guided acupotomy release combined with low-level He-Ne laser therapy group (ALG), ultrasound-guided acupotomy release combined with corticosteroid injection group (ACG), or ultrasound-guided acupotomy release group (AG).

Baseline demographic and clinical characteristics were collected and compared among groups to evaluate intergroup comparability. Clinical outcomes, including pain intensity, functional status, electrophysiological findings, imaging parameters, and adverse events, were retrospectively analyzed before and after treatment.

Patients were included if they met the following criteria: (1) Age≥18 years; (2) Clinical symptoms consistent with CTS, including hand pain, numbness, paresthesia, or nocturnal symptoms in the median nerve distribution; (3) CTS diagnosis confirmed by electrophysiological examination; (4) Mild-to-moderate CTS according to clinical and electrophysiological findings; (5) Received one of the study interventions and completed follow-up evaluations. Patients were excluded if they had: (1) Previous wrist surgery or local injection treatment within the past 6 months; (2) Severe CTS requiring immediate surgical decompression; (3) Patients who received regular physiotherapy (including ultrasound

therapy, shock-wave therapy, manual therapy or nerve-gliding training) within 3 months before enrolment; (4) Cervical radiculopathy, peripheral neuropathy, or other neurological disorders affecting upper limb symptoms; (5) Space-occupying lesions within the carpal tunnel; (6) Severe systemic disease, active infection, coagulation disorders, or malignancy; (7) Pregnancy or lactation; (8) Incomplete clinical or follow-up data.

Intervention Methods

Ultrasound-Guided Acupotomy Release

After routine disinfection of the affected wrist under ultrasound guidance, a puncture needle was first inserted to the junction of the transverse carpal ligament and the median nerve, and 5 mL of 1% lidocaine hydrochloride was injected to separate the two structures (Figure 1A). Subsequently, 1 mL of 1% lidocaine hydrochloride was injected onto the surface of the transverse carpal ligament (Supplementary Video 1). After anesthesia took effect, a 1.2 mm×50 mm acupotomy (Jiangxi Laozongyi Medical Instrument Co., Ltd.) was used, with the blade direction aligned with the course of the median nerve and the acupotomy forming a 15°angle with the skin. Under ultrasound guidance, the acupotomy was inserted from the proximal edge of the transverse carpal ligament (Figure 1B) and advanced from superficial to deep layers for gradual release. The release procedure was maintained along a straight line until the resistance during pushing and lifting nearly disappeared, and ultrasound revealed a marked reduction in the echogenicity of the transverse carpal ligament (Supplementary Videos 2 and 3). The acupotomy was then withdrawn, and sterile gauze was applied with compression. Postoperatively, the patient's wrist was wrapped with an elastic bandage.

Ultrasound-Guided Acupotomy Release Combined with Corticosteroid Injection

The patient positioning and disinfection method were the same as those used in AG. Prior to acupotomy treatment, a drug mixture consisting of 2 mL of 2% lidocaine hydrochloride injection, 3 mL of 0.9% NaCl injection, and 1 mL of compound betamethasone injection (Schering-Plough Labo N.V.) was administered. The specific injection technique and the acupotomy release procedure were identical to those described for the AG.

Ultrasound-Guided Acupotomy Release Combined with Low-Level He-Ne Laser Therapy

The preoperative anesthesia and ultrasound-guided acupotomy release procedure were the same as those used in Group AG. After completion of the acupotomy release, a disposable laser acupotomy needle (1.0 mm × 50 mm, Hefei Yongkang Electronic Technology Co., Ltd.) was inserted under ultrasound guidance to the junction of the transverse carpal ligament and the median nerve (Figure 2A and B). The laser acupotomy needle was then connected to an SJ-L1 laser acupotomy therapeutic apparatus (Hefei Yongkang Electronic Technology Co., Ltd.). The He-Ne laser produces red visible light with a wavelength of 632.8 nm. The output power was set at 18 mW, with irradiation duration of 5 min. The total delivered



Figure 1 (**A**) Ultrasound-guided hydrodissection of the median nerve; (**B**) Ultrasound-guided acupotomy release of the transverse carpal ligament.

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Figure 2 (**A**) a: laser acupotomy needle; b: median nerve in long axis; c: transverse carpal ligament; (**B**) Ultrasound-guided insertion of the laser acupotomy needle; (**C**) Image after connection to the laser therapeutic apparatus, showing red visible light emission at the site of the transverse carpal ligament.

energy was calculated as 5.4 J, and the energy density at target tissue was 12.2 J/cm^2 (Figure 2C). After irradiation, the laser acupotomy needle was withdrawn, and sterile gauze was applied with compression. Postoperatively, the patient's wrist was wrapped with an elastic bandage.

Evaluation Indicators

The primary outcome of this study was improvement in pain intensity, which was assessed using the Visual Analog Scale (VAS).[21] VAS scores were recorded before treatment and at 1, 4, and 16 weeks after treatment.

Secondary outcomes included functional recovery, electrophysiological findings, ultrasonographic parameters, and treatment-related adverse events.

Functional outcomes were evaluated using the Boston Carpal Tunnel Questionnaire (BCTQ).[22] Higher scores indicated greater symptom severity and functional impairment.

Electrophysiological assessment of the median nerve was performed using standard nerve conduction studies, including Motor conduction parameters included latency (Lat), amplitude (Amp), and motor conduction velocity (MCV). Sensory conduction parameters included Lat, Amp, and sensory conduction velocity (SCV).

Ultrasonographic evaluation was conducted using high-frequency ultrasound and shear wave elastography. Parameters included the shear wave elasticity values of the transverse carpal ligament and the flattening ratio (FR) of the median nerve (Figure 3A–D).

Follow-Up

An online follow-up was conducted 4 months after treatment to record any adverse events during this period, including nerve injury, muscle atrophy, and infection. Disease recurrence and cases requiring surgical intervention due to ineffective treatment were also documented.

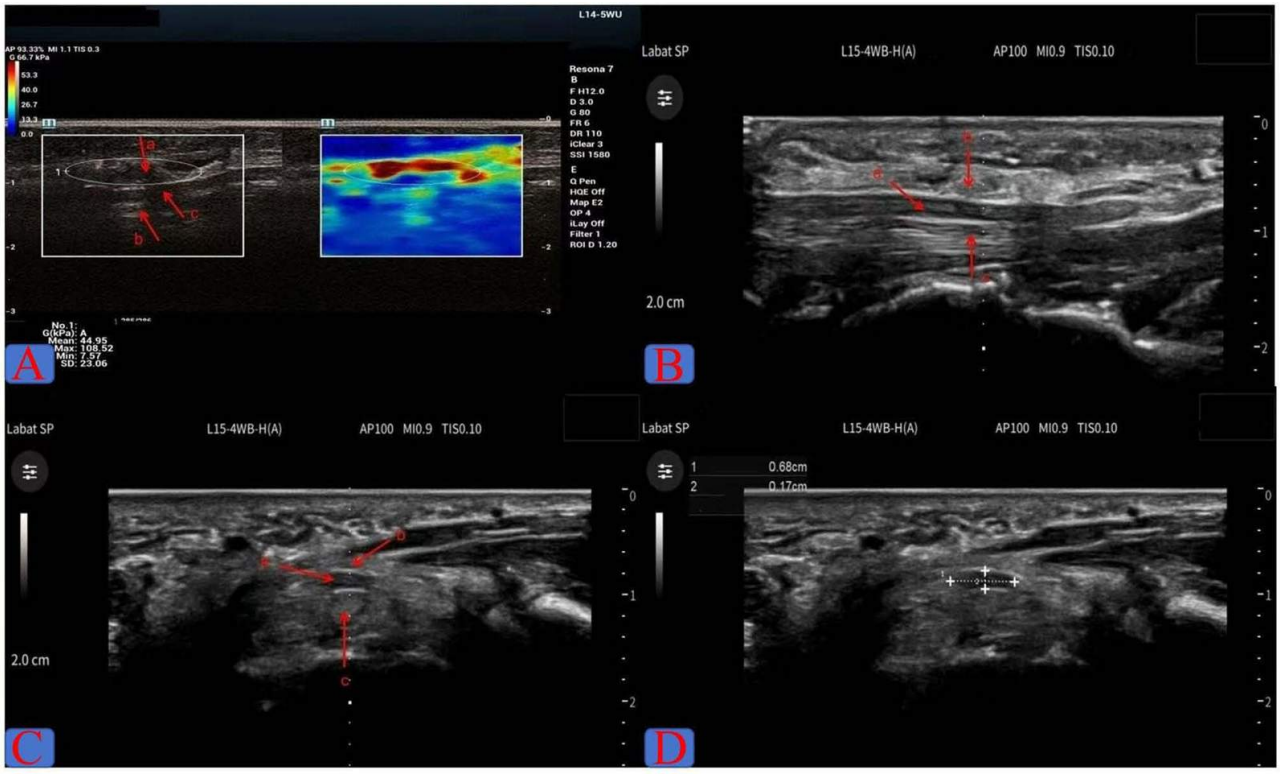


Figure 3 (**A**) Shear wave elastography of the transverse carpal ligament. (a: transverse carpal ligament; b: flexor tendons; c: median nerve in long axis). The deep red area indicates higher elasticity values, while the blue area indicates lower elasticity values; (**B**) Long-axis view of the median nerve. (a: median nerve in long axis; b: transverse carpal ligament; c: flexor tendons); (**C**) Short-axis view of the median nerve. (a: median nerve in short axis; b: transverse carpal ligament; c: flexor tendons); (**D**) Measurement of the flattening ratio of the compressed median nerve in short-axis view.

Statistical Analysis

Statistical analyses were performed using SPSS 26.0 software (IBM Inc., Chicago, IL). Continuous variables were expressed as mean±standard deviation (SD), whereas categorical variables were presented as frequencies and percentages. Baseline demographic and clinical characteristics were compared among groups to evaluate intergroup comparability. Continuous variables were analyzed using one-way analysis of variance (ANOVA) or the Kruskal–Wallis test, as appropriate, while categorical variables were analyzed using the chi-square test or Fisher's exact test. Changes in clinical outcomes over time were analyzed using repeated-measures analysis of variance. Post hoc pairwise comparisons were performed using Bonferroni correction when significant differences were identified. The primary outcome was improvement in Visual Analog Scale (VAS) scores. Secondary outcomes included BCTQ scores, electrophysiological findings, ultrasonographic parameters, and treatment-related adverse events. To reduce potential confounding associated with the retrospective non-randomized study design, multivariable regression analysis was performed with adjustment for baseline covariates, including age, sex, diabetes status, and symptom duration. Additionally, analysis of covariance (ANCOVA) was performed for the primary outcome VAS score at each follow-up time-point, with baseline VAS score set as the covariate, to compare post-treatment pain outcomes across three groups while controlling for baseline pain levels. Patients with incomplete follow-up or missing outcome data were excluded from the final analysis. A two-sided P value<0.05 was considered statistically significant.

Result

General Demographics

A total of 201 patients with CTS were initially enrolled. Among them, 15 were excluded due to bilateral CTS, 2 for previous CTS-related surgery, and 2 due to severe thenar muscle atrophy. Thus, 182 patients completed the treatment and were assigned to the following groups based on treatment received: ALG (n=58), ACG (n=62), and AG (n=62). During follow-up, 1 patient in ALG was excluded for self-administered pain medication, 2 were lost to follow-up, and 1

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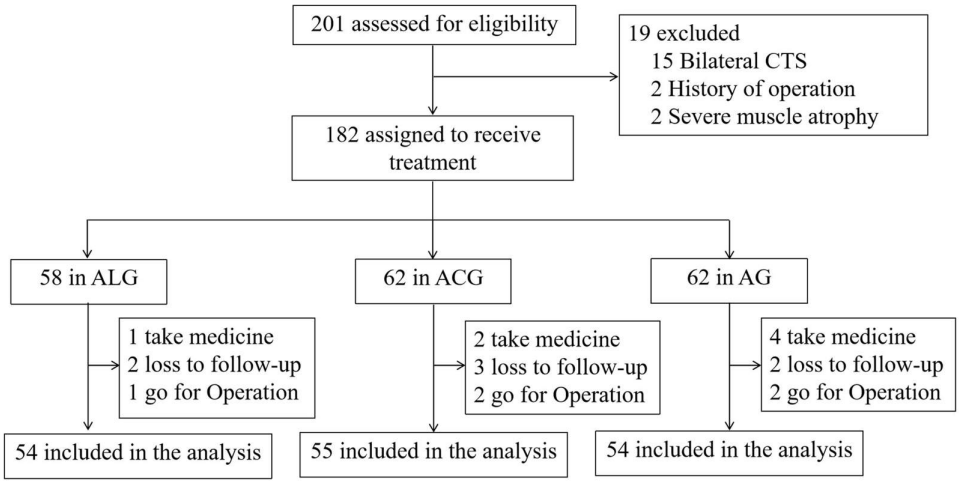


Figure 4 Flow diagram of the study.

underwent surgical treatment. In ACG, 2 were excluded for self-medication, 3 were lost, and 2 sought surgery. In AG, 4 self-medicated, 2 were lost, and 2 received surgery. Consequently, the final analysis included 54 patients in ALG, 55 in ACG, and 54 in AG (Figure 4). No significant differences were observed among the three groups in baseline characteristics such as age, disease duration, or electrophysiological results ($P > 0.05$). See Table 1. The balanced

Table 1 Comparison of Baseline Data Among 3 Groups

Characteristics	ALG (n=54)	ACG (n=55)	AG (n=54)	*P* value
Gender, n (%)				0.562
Male	12(22.22)	7(12.73)	9(16.67)	
Female	42(77.78)	48(87.27)	45(83.33)	
Age (mean±SD, years)	53.02±10.68	51.47±10.64	54.98±7.83	0.175
Course of disease (mean±SD, months)	5.26±2.03	5.44±1.89	5.19±1.93	0.789
Affected side, n (%)				0.636
Right	29(53.70)	34(61.82)	33(61.11)	
Left	25(46.30)	21(38.18)	21(38.89)	
Hand dominance, n (%)				0.931
Left	7(12.96)	6(10.91)	7(12.96)	
Right	47(87.04)	49(89.09)	47(87.04)	
Engage in manual labor, n (%)				0.295
Yes	37(68.52)	30(54.55)	35(64.81)	
No	17(31.48)	25(45.45)	19(35.19)	
Comorbidities, n (%)				
Diabetes	8(14.81)	13(23.64)	12(22.22)	0.470
Hypertension	13(24.07)	11(20.00)	15(27.78)	0.636
Motor Conduction				
Lat, (mean±SD, ms)	5.36±0.14	5.36±0.16	5.31±0.12	0.148
Amp, (mean±SD, mV)	4.56±0.22	4.52±0.14	4.52±0.14	0.412
MCV, (mean±SD, m/s)	40.90±2.55	41.04±1.91	42.27±2.38	0.712
Sensory Conduction				
Lat, (mean±SD, ms)	4.42±0.17	4.39±0.13	4.37±0.21	0.285
Amp, (mean±SD, mV)	15.62±0.99	15.67±1.69	15.73±1.61	0.928
SCV, (mean±SD, m/s)	42.09±3.45	43.24±2.78	43.02±2.41	0.103

Abbreviations: ALG, Acupotomy release and low energy He-Ne laser group; ACG, Acupotomy release and corticosteroid injection group; AG, Acupotomy release group; Lat, Latency; Amp, Amplitude; MCV, Motor Conduction Velocity; SCV, Sensory Conduction Velocity; SD, Standard Deviation.

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baseline characteristics among the three groups further confirmed that the non-random grouping based on patient choice and clinical evaluation did not introduce significant selection bias, which ensured the comparability of the therapeutic effects among the groups.

Multivariable Regression Analysis

Multivariable regression analysis was performed to adjust for potential baseline confounders, including age, sex, diabetes mellitus, and symptom duration. After adjustment, both the ALG (β = 0.901, 95% CI: 0.587–1.215, P<0.001) and the ACG (β = 0.842, 95% CI: 0.525–1.158, P<0.001) remained independently associated with greater improvement in VAS scores compared with the AG. No significant associations were identified for age, sex, diabetes mellitus, or symptom duration. Detailed results are presented in Table 2.

VAS Scores

VAS scores significantly decreased in all three groups after treatment (P<0.05). Compared with the AG, both the ALG and ACG demonstrated significantly greater pain reduction at 1, 4, and 16 weeks after treatment (P<0.05 for all comparisons). No significant differences in VAS scores were observed between the ALG and ACG throughout follow-up, indicating that low-level He-Ne laser therapy achieved pain relief comparable to corticosteroid injection. These findings were further supported by ANCOVA adjusting for baseline VAS values, which yielded consistent inter-group comparisons. Detailed results are provided in Table 3 and Figure 5.

BCTQ Scores

After treatment, the Boston Carpal Tunnel Questionnaire (BCTQ) scores decreased significantly in all three groups ($P < 0.05$). Compared with the AG group, both the ALG and ACG showed significantly lower BCTQ scores at 1 week, 4 weeks, and 16 weeks post-treatment ($P < 0.05$). Furthermore, the ACG exhibited significantly lower BCTQ scores than the ALG at both 1 week and 4 weeks after treatment ($P < 0.05$). These findings suggest that both low-level He-Ne laser

Table 2 Multivariable Regression Analysis for Improvement in VAS Scores

Variable	*β* Coefficient	*95% CI*	*P* value
ACG	0.842	0.525~1.158	<0.001
ALG	0.901	0.587~1.215	<0.001
Age	−0.003	−0.018~0.011	0.645
Female sex	−0.022	−0.364~0.319	0.898
Diabetes mellitus	0.020	−0.335~0.375	0.912
Symptom duration	0.0004	−0.066~0.066	0.990

Abbreviations: ALG, Acupotomy release and low energy He-Ne laser group; ACG, Acupotomy release and corticosteroid injection group.

Table 3 Comparison of VAS Scores Among the Three Groups [$M(P_{25},P_{75})$, Scores]

Group	Baseline	1 Week	4 Weeks	16 Weeks
ALG (n=54)	7(6,7)	$3(3,4)^{\#*}$	$1(1,2)^{\#*}$	$1(0,1)^{\#*}$
ACG (n=55)	7(6,7)	$3(2,3)^{\#*}$	$1(1,2)^{\#*}$	$1(0,1)^{\#*}$
AG (n=54)	7(6,7)	$4(4,5)^{\#}$	$3(2,3)^{\#}$	$1(1,2)^{\#}$
P value	0.990	<0.001	<0.001	<0.001

Notes: Compared to Baseline, $^{\#}P<0.05$; Compared to AG, $^{*}P<0.05$.
Abbreviations: ALG, Acupotomy release and low energy He-Ne laser group; ACG, Acupotomy release and corticosteroid injection group; AG, Acupotomy release group.

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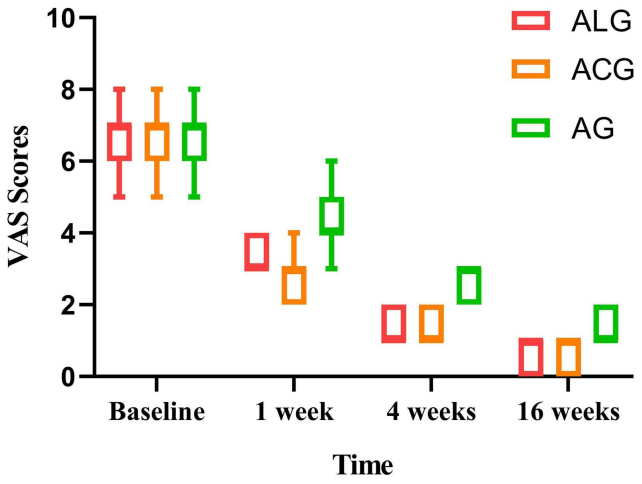


Figure 5 Comparison of VAS scores among the three groups.

therapy and corticosteroid injection contributed to superior functional recovery compared with acupotomy alone. Detailed results are provided in Table 4 and Figure 6.

Electrophysiologica Results

Electrophysiological parameters improved significantly in all groups after treatment. Compared with the AG, both the ALG and ACG demonstrated greater improvement in median nerve conduction function ($p<0.05$). Detailed results are presented in Table 5 and Figure 7.

Imaging Outcomes

Imaging outcomes improved significantly in all groups after treatment. Compared with the AG, both the ALG and ACG demonstrated greater improvement in FR of the median nerve and Young's Modulus ($p<0.05$). Detailed results are presented in Table 6 and Figure 8.

Adverse Events

No serious adverse events, including median nerve injury or infection, were observed in any group. Steroid-related hyperglycemia occurred in six patients in the ACG and resolved after glucose monitoring and medication adjustment. In contrast, no laser-related adverse events were identified in the ALG during follow-up. These findings support the favorable safety profile of low-level He-Ne laser therapy as a steroid-sparing adjunctive treatment strategy for CTS.

Table 4 Comparison of BCTQ Scores Among 3 Groups(mean±SD, Scores)

Group	Baseline	1 Week	4 Weeks	16 Weeks
ALG (n=54)	70.17±5.55	46.63±3.94$^{\#*}$	32.50±3.98$^{\#*}$	22.78±2.11$^{\#*}$
ACG (n=55)	71.02±4.62	41.91±4.27$^{\#*\wedge}$	30.24±3.72$^{\#*\wedge}$	22.51±2.12$^{\#*}$
AG (n=54)	71.31±3.39	52.98±4.49$^{\#}$	40.76±4.90$^{\#}$	29.37±2.74$^{\#}$
P value	0.413	<0.001	<0.001	<0.001

Notes: Compared to Baseline, $^{\#}P<0.05$; Compared to AG, $^{*}P<0.05$; Compared to ALG, $^{\wedge}P<0.05$.

Abbreviations: ALG, Acupotomy release and low energy He-Ne laser group; ACG, Acupotomy release and corticosteroid injection group; AG, Acupotomy release group.

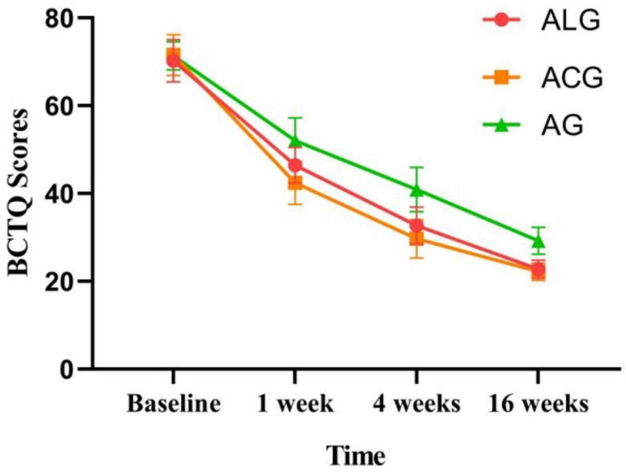


Figure 6 Comparison of BCTQ scores among 3 groups.

Discussion

The present multicenter retrospective study demonstrated that ultrasound-guided low-level He-Ne laser therapy combined with acupotomy release significantly improved pain intensity, functional outcomes, electrophysiological findings, and ultrasonographic parameters in patients with CTS. Importantly, the therapeutic efficacy of the ALG was generally comparable to that of corticosteroid injection, while avoiding steroid-related adverse events such as hyperglycemia. These findings suggest that low-level He-Ne laser therapy may represent a clinically feasible steroid-sparing adjunctive strategy for CTS management.

Pain relief and functional recovery are major therapeutic goals in CTS management. In the present study, both the ALG and ACG demonstrated significantly greater improvement in VAS and BCTQ scores compared with acupotomy alone. These findings indicate that adjunctive therapies following minimally invasive decompression may further enhance symptomatic recovery. Notably, the comparable outcomes observed between the ALG and ACG suggest that low-level He-Ne laser therapy may provide analgesic and functional benefits similar to corticosteroid injection.

Table 5 Comparison of Electrophysiologica Results Among 3 Groups

Electrophysiologica Index	Time	ALG (n=54)	ACG (n=55)	AG (n=54)	*P* value
Motor Conduction					
Lat, (mean±SD, ms)	Before Treatment	5.36±0.14	5.36±0.16	5.31±0.12	0.148
	After Treatment	3.04±0.14$^{\#*}$	3.05±0.14$^{\#*}$	3.97±0.21$^{\#}$	<0.001
Amp, (mean±SD, mV)	Before Treatment	4.56±0.22	4.52±0.14	4.52±0.14	0.412
	After Treatment	10.22±0.47$^{\#*}$	10.53±0.49$^{\#*}$	8.94±0.33$^{\#}$	<0.001
MCV (mean±SD, m/s)	Before Treatment	40.90±2.55	41.04±1.91	42.27±2.38	0.712
	After Treatment	57.98±3.87$^{\#*}$	57.58±3.67$^{\#*}$	51.27±3.58$^{\#}$	<0.001
Sensory Conduction					
Lat, (mean±SD, ms)	Before Treatment	4.42±0.17	4.39±0.13	4.37±0.21	0.285
	After Treatment	2.14±0.13$^{\#*}$	2.07±0.14$^{\#*}$	3.10±0.13$^{\#}$	<0.001
Amp, (mean±SD, mV)	Before Treatment	15.62±0.99	15.67±1.69	15.73±1.61	0.928
	After Treatment	28.29±2.48$^{\#*}$	28.40±2.45$^{\#*}$	21.66±1.77$^{\#}$	<0.001
SCV (mean±SD, m/s)	Before Treatment	42.09±3.45	43.24±2.78	43.02±2.41	0.103
	After Treatment	68.39±6.22$^{\#*}$	68.92±5.39$^{\#*}$	56.12±3.39$^{\#}$	<0.001

Notes: Compared to Before treatment, $^{\#}P<0.05$; Compared to AG, $^{*}P<0.05$.

Abbreviations: ALG, Acupotomy release and low energy He-Ne laser group; ACG, Acupotomy release and corticosteroid injection group; AG, Acupotomy release group; Lat, Latency; Amp, Amplitude; MCV, Motor Conduction Velocity; SCV, Sensory Conduction Velocity; SD, Standard Deviation.

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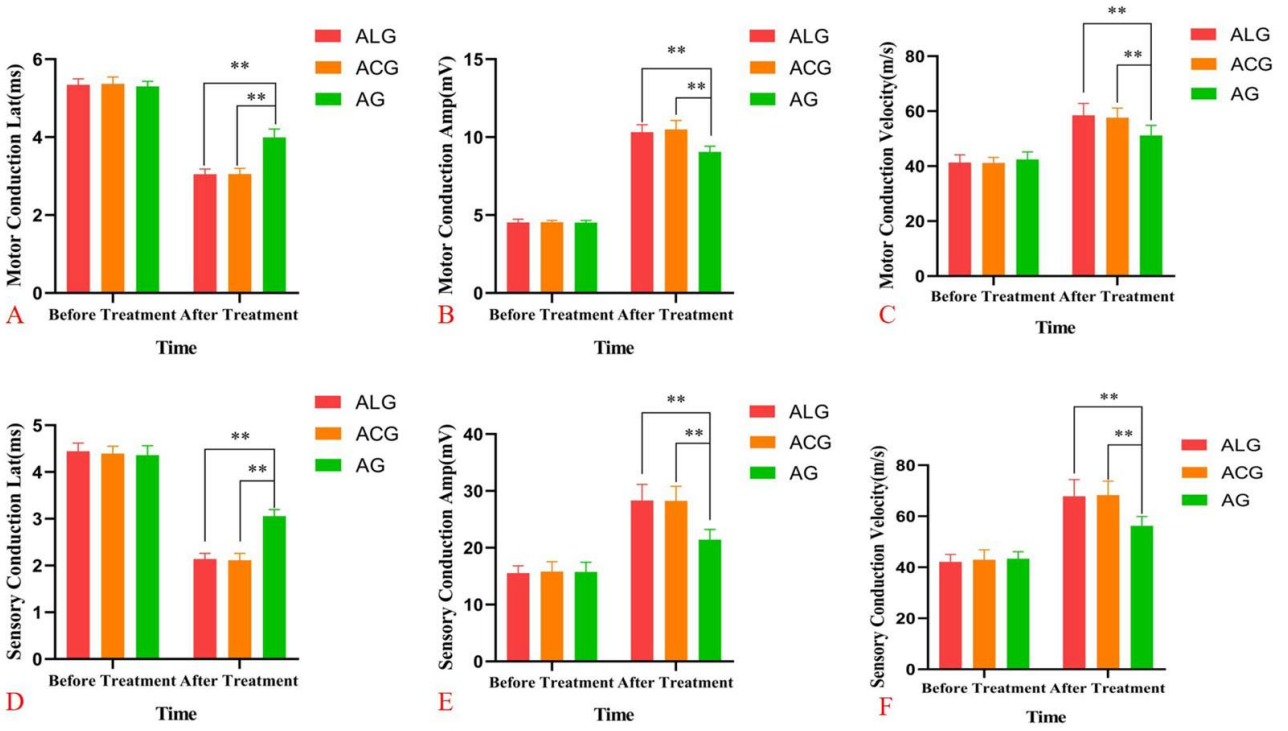


Figure 7 Changes in nerve electrophysiological indicators before and after treatment in each group. (**A**) Motor conduction latency; (**B**) Motor conduction amplitude; (**C**) Motor conduction velocity; (**D**) Sensory conduction latency; (**E**) Sensory conduction amplitude; (**F**) Sensory conduction velocity. ALG, ACG and AG represent different experimental groups. (**P<0.001).

Low-level laser therapy (LLLT) has been widely applied in the treatment of various musculoskeletal and neurological disorders.[23] Unlike high-energy lasers that produce thermal ablation, LLLT exerts non-thermal photobiomodulatory effects through low-power monochromatic or quasi-monochromatic light.[24,25] After photons are absorbed by mitochondrial chromophores such as cytochrome c oxidase, intracellular ATP synthesis, nitric oxide release, and microcirculatory activity are enhanced,[26,27] thereby producing anti-inflammatory, analgesic, tissue repair, and anti-fibrotic effects.[28,29] Among different low-level lasers, He-Ne laser has been widely used because of its favorable biocompatibility and established clinical efficacy.[30] Its red visible wavelength is particularly suitable for soft tissue disorders due to its relatively efficient tissue absorption characteristics.[31]

In CTS, chronic compression of the median nerve beneath the transverse carpal ligament results in local microcirculatory disturbance, inflammatory edema, and fibrosis.[32] Low-level He-Ne laser therapy may alleviate these pathological changes by suppressing inflammatory mediator release, promoting neural metabolism and axoplasmic transport,

Table 6 Comparison of Imaging Outcomes Among 3 Groups

Imaging Index	Time	ALG (n=54)	ACG (n=55)	AG (n=54)	*P* value
Transverse diameter (mean±SD, mm)	Before Treatment	7.02±0.69	7.01±0.75	7.07±0.82	0.888
	After Treatment	5.15±0.44$^{\#*}$	5.15±0.63$^{\#*}$	6.06±0.42$^{\#}$	<0.001
Anteroposterior diameter (mean±SD, mm)	Before Treatment	1.73±0.18	1.72±0.15	1.72±0.17	0.878
	After Treatment	2.35±0.16$^{\#*}$	2.40±0.21$^{\#*}$	2.10±0.10$^{\#}$	<0.001
FR (mean±SD)	Before Treatment	3.90±0.56	3.98±0.60	3.97±0.73	0.764
	After Treatment	2.10±0.32$^{\#*}$	2.19±0.27$^{\#*}$	2.98±0.25$^{\#}$	<0.001
Young's Modulus (mean±SD, kPa)	Before Treatment	44.63±3.89	43.74±4.39	43.20±4.03	0.200
	After Treatment	24.44±2.64$^{\#*}$	25.45±3.49$^{\#*}$	31.99±3.26$^{\#}$	<0.001

Notes: Compared to Before treatment, $^{\#}P$<0.05; Compared to AG, $^{*}P$<0.05.
Abbreviations: ALG, Acupotomy release and low energy He-Ne laser group; ACG, Acupotomy release and corticosteroid injection group; AG, Acupotomy release group; SD, Standard Deviation; FR, Flattening Ratio.

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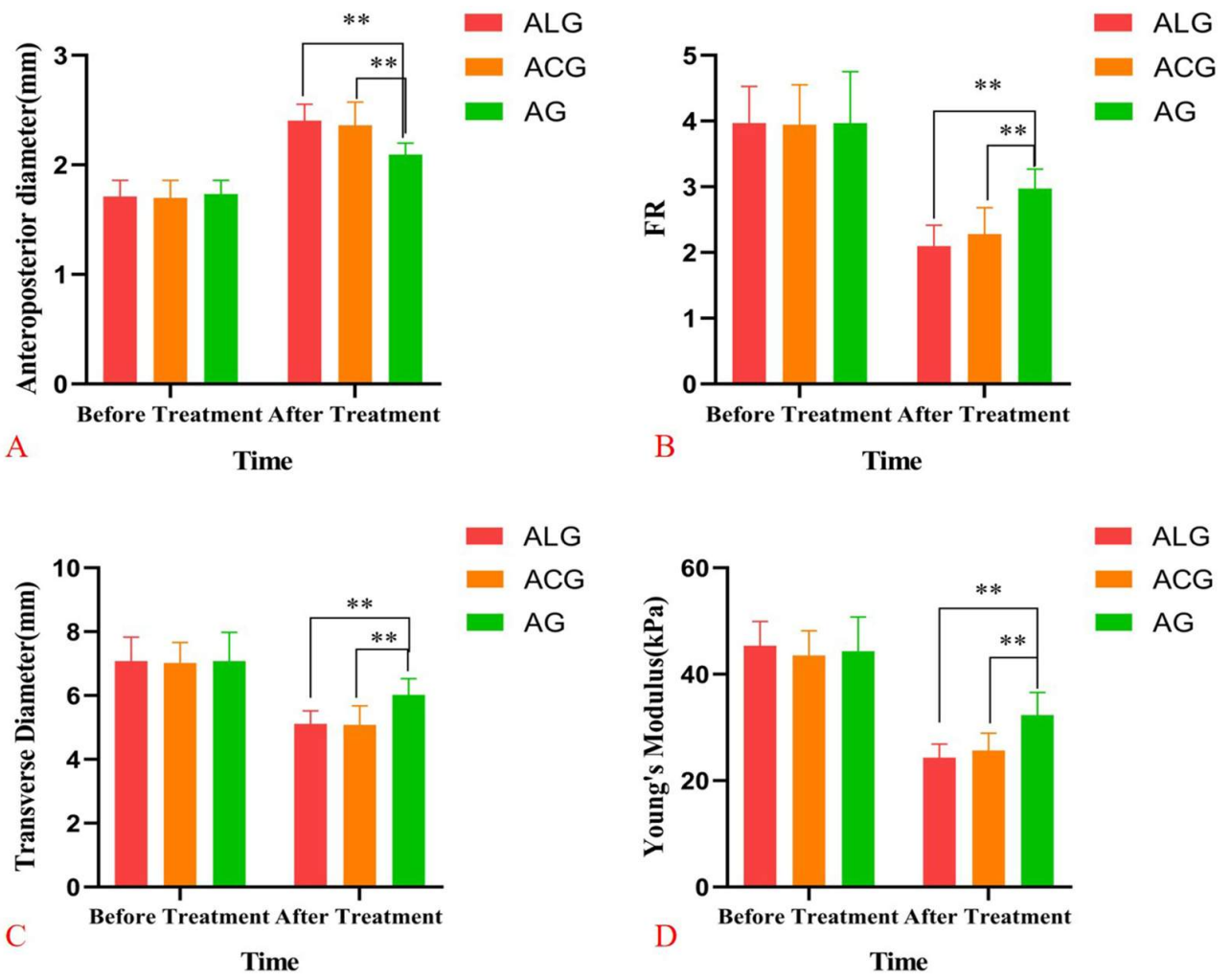


Figure 8 Changes in nerve morphological and biomechanical parameters before and after treatment in each group. (**A**) Anteroposterior diameter; (**B**) Friction coefficient (FR); (**C**) Transverse diameter; (**D**) Young's modulus. ALG, ACG and AG represent different experimental groups. (**$P < 0.001$).

and modulating fibroblast activity.[33–35] These mechanisms may contribute to pain relief, nerve recovery, and reduction of postoperative scar adhesion following acupotomy release. Ghasemi et al also applied LLLT to patients with CTS, and the results demonstrated that both the cross-sectional area of the median nerve and pain symptoms were improved to varying degrees after treatment, which is consistent with the findings of the present study.[36]

However, the clinical efficacy of conventional transcutaneous He-Ne laser therapy is limited by its shallow tissue penetration, as the wavelength of approximately 632.8 nm can only penetrate several millimeters into biological tissues.[37] In CTS, the principal pathological region is located beneath the transverse carpal ligament, where the compressed median nerve, inflamed synovium, and fibrotic tissues are deeply situated beneath the skin and fascia. Consequently, a substantial proportion of laser energy is absorbed or scattered within superficial tissues during transcutaneous irradiation, reducing the effective energy delivered to the target region. In contrast, the present study utilized ultrasound-guided targeted delivery of He-Ne laser therapy following acupotomy release, which may facilitate direct photobiomodulation at the site of median nerve compression. This targeted intervention strategy may contribute to local anti-inflammatory effects, microcirculatory improvement, and neural recovery, thereby enhancing treatment efficacy.

Corticosteroid injection remains a commonly used treatment for CTS. However, steroid-related adverse effects remain a clinical concern, particularly in patients with diabetes mellitus and metabolic comorbidities. In the present study, steroid-associated hyperglycemia occurred only in the ACG, whereas no laser-related systemic adverse events

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were observed in the ALG. These findings support the favorable safety profile of low-level He-Ne laser therapy and highlight its potential value as a steroid-sparing treatment option for CTS patients at increased metabolic risk.

The strengths of this study include its multicenter design, comprehensive clinical and objective outcome assessments, and adjustment for baseline confounding factors using multivariable regression analysis. In addition, the use of ultrasound-guided targeted intervention improved procedural accuracy and treatment reproducibility. Several limitations should be acknowledged. First, this was a retrospective non-randomized study, which may have introduced selection bias despite multivariable adjustment. Second, the follow-up duration was relatively short, limiting evaluation of long-term therapeutic efficacy. Third, the sample size was moderate, and larger prospective randomized studies are warranted to further validate these findings. In addition, we did not perform predefined age-stratified subgroup comparisons. Subgroup stratification would subdivide the moderate-size overall sample and result in small patient numbers within each age stratum, potentially introducing low statistical power. Future larger-sample prospective studies are warranted to explore potential age-related differences in treatment response.

Conclusions

In conclusion, ultrasound-guided low-level He-Ne laser therapy combined with acupotomy release yields favorable pain-relieving effects, functional improvements, and safety profiles in patients with CTS. This intervention achieves therapeutic efficacy generally comparable to corticosteroid injection, yet without steroid-related adverse events. This targeted photobiomodulation strategy may serve as a promising minimally invasive steroid-sparing approach for the management of CTS.

Data Sharing Statement

Whether the authors intend to share individual deidentified participant data: Yes.

What specific data they intend to share: Patient rating scale results.

What other study documents will be made available: Imaging information during patient treatment process.

How the data will be accessible (publishable contact details): According to the corresponding author of this study: Yong-Hui Yang, email: yangyonghui@ahtcm.edu.cn

When and for how long they will be made available: The data will be obtained 3 years after the publication of the article.

Author Contributions

All authors made a significant contribution to the work reported, whether that is in the conception, study design, execution, acquisition of data, analysis and interpretation, or in all these areas; took part in drafting, revising or critically reviewing the article; gave final approval of the version to be published; have agreed on the journal to which the article has been submitted; and agree to be accountable for all aspects of the work.

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Disclosure

The authors declared that they have no conflicts of interest regarding this work.

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